EXECUTIVE SUMMARY AND BUSINESS RECOMMENDATIONS

Healthcare Operations Dashboard Overview

I analysed 27,890 encounters from the healthcare operations dataset. Total claim cost was $101.51M, but payer coverage was only $31.10M, leaving a large gap. Two issues stood out: almost half of all visits had zero payer coverage, and a small group of patients stayed longer than 24 hours, which is expensive and blocks beds.

What the Dashboard Showed

- 13,586 encounters (48.7%) had no payer coverage at all. This was most common in wellness and ambulatory visits.
- Only 76 encounters (0.27%) went over 24 hours, but they were all inpatient and cost far more than average.
- There is a $70M difference between what we billed and what payers covered.
- Emergency is being used a lot for cases that could be handled elsewhere.

Recommendation 1: Fix Insurance Checks Before the Patient Arrives

Finding: 48.7% of our visits show PAYER_COVERAGE = 0. When I filtered by encounter class, wellness visits were the worst affected. This suggests we are not checking eligibility properly, or patients are marked as self-pay by default.

Action: I recommend adding an automatic eligibility check 24 hours before the appointment, especially for wellness visits. Front desk should re-check on arrival and always ask for a secondary payer. We can track this weekly using the Zero Coverage KPI I built.

Benefit: If we can reduce zero-coverage visits by just 20%, that is about 2,717 encounters. At an average claim of $3,639, we could recover roughly $9.8M that is currently being lost to bad debt.

Limitation: This will need a clearinghouse integration and staff training. Also, this is synthetic data, and in real life some patients will still be uninsured and stay at zero coverage.

Recommendation 2: Manage Long-Stay Patients Better

Finding: Only 76 cases were over 24 hours, but they tie up beds for days. These are inpatient cases and they drive our highest cost per encounter.

Action: Set up a daily discharge huddle. Any patient approaching 20 hours should trigger an alert in Power BI — I already have the Duration_Hours measure for this. The team should check what is holding up discharge: labs, consults, or transport.

Benefit: Cutting the average long-stay from 48 hours to 36 hours would free up over 900 bed-hours a year. That improves patient flow and lowers the risk of complications.

Limitation: We cannot rush clinical care. Some patients genuinely need to stay longer. This also affects a very small group (0.27%), so the total cost saving is not huge, but the operational impact is important.

Recommendation 3: Move Low-Acuity Cases Out of Emergency

Finding: Emergency has a high cost and low coverage compared to urgent care and outpatient. The trend page shows consistent emergency volume that looks like primary care issues ending up in the ED.

Action: Use the dashboard to pull the top 10 low-acuity codes that keep appearing in emergency. Use that list to educate patients and to create a triage guideline so frontline staff can redirect suitable cases to urgent care. Extend urgent care hours slightly to absorb this.

Benefit: Urgent care costs about 40% less than emergency. If we shift even 10% of emergency cases (around 300 visits), we save about $450K per year and we decongest the ED, which improves wait times.

Limitation: We have to follow EMTALA — we cannot turn anyone away from the ED. Patient behaviour is also hard to change quickly, and we would need to staff urgent care for longer hours.

Final Conclusion

The dashboard highlights three clear opportunities: recover revenue from zero-coverage visits, improve bed availability by managing long stays, and lower costs by shifting the right patients to urgent care. All three can be tracked directly in the dashboard using the measures I created — Zero Coverage (13,586), % Over 24h (0.27%), and Cost by Encounter Class.